

BACKTRACK: validation before submission

What the application still needs

The software and proposal are not evidence that students need or will use BACKTRACK. Before September 18, seek 10–15 permitted learner interviews/walkthroughs and 3–5 educator interviews. These are recruitment targets, not completed interviews. A small convenience sample can identify problems but cannot estimate national demand.

The team must arrange school and adult permissions for any activity involving minors. Do not collect identifiable student assessment screenshots, photos, or quotes without the required permissions. The scripts below are ready for the team to use; no outreach has been sent on its behalf.

Recruitment

Recruit learners with varied recent experiences of mathematics: those who felt stuck, those who used AI or Khan to catch up, and those who already cope well. Avoid recruiting only friends who know the pitch. Invite teachers who handle current secondary topics, remedial support, and shared-device settings. Include at least one skeptical educator who can identify incorrect prerequisite assumptions.

Record how each participant was recruited, their permission status, broad level/context, and whether they had previously seen the product. Do not record unnecessary personal information in the research sheet.

Learner interview: 12 minutes

Opening: “We’re testing the tool, not you. You can skip anything or stop. Honest criticism helps us fix it.”

1. “Tell me about the last mathematics lesson that stopped making sense.” Ask for a specific recent example.
2. “What did you do next?” Ask what they actually did before asking what they might do.
3. “How did you decide what to review?” Ask what was difficult about choosing the earlier topic.
4. “Did you use Khan, a teacher, notes, a friend, or an AI tutor? What still did not help?”
5. “What access did you have: your own phone, a shared device, data, or school internet?”
6. “When you return after a break, what makes starting hard?”

Do not ask whether they “want a GPS for learning” before hearing the problem. Do not imply they should have used Khan.

Learner walkthrough: 10 minutes

Show the homepage without a speech. After ten seconds ask, “What would you do here?” Then ask them to choose a destination, start a short block, answer honestly, and reach a learning stop. Avoid coaching unless they cannot continue; record each intervention.

Observe whether they can name the destination, understand the task, use the controls, interpret a route change, find Khan practice, return to a fresh check, and pause. Ask them to explain “Shortcut found” in their own words. Do not equate saying “cool” with understanding.

At the optional continuation, ask them to choose naturally. Then ask why. Ask “Did any part make you feel rushed, judged, or pressured?” and “What would make you not use this?” Finish by asking which single screen they would remove or change.

Teacher interview: 20 minutes

1. “Which current topic most often reveals earlier gaps? Can you show an anonymized example?”
2. “How do you work out the blocker now, and how long does that take?”
3. “Which dependency in this destination map is wrong, missing, or optional?”
4. “Could a learner answer this item correctly using a different method?”
5. “What would count as returning to today’s lesson? Review forms A/B/C and the rubric.”
6. “Where could practice realistically fit without adding an unsustainable task?”
7. “What Khan classes, reports, or exports are you actually authorized to use?”
8. “Which learners would be excluded by the proposed access arrangement?”
9. “What would make you refuse or stop using this?”
10. “Who can authorize a small cycle? What would they need to see first?”

Ask separately about permission to describe the interview, quote them, name the institution, or describe any future participation. One permission does not imply the others.

Data schema

Use one row per session with these columns: participant_code, participant_type, recruitment_channel, broad_learning_context, prior_exposure, permission_status, date, interviewer, recent_problem, current_workaround, access_barrier, destination_understood, first_action, task_completion, assistance_count, route_explanation, khan_found, return_found, next_turn_choice, pressure_rating, stop_reason, observed_issue, permitted_quote, quote_permission_scope, proposed_change, evidence_file_reference.

For educator review, add destination_id, prerequisite_issue, alternate_method, assessment_issue, schedule_option, report_access_status, acceptable_burden, decision_owner, interest_status, naming_permission, follow_up_owner.

Keep interview identifiers and consent records in the institution's approved restricted location. This schema contains no actual participant data.

Decision rules

- If several learners cannot explain the destination after ten seconds, revise the first screen before adding features.
- If learners cannot distinguish a resource open from completed practice, revise the Khan checkpoint and evidence labels.
- If a teacher identifies a false dependency or valid rejected answer, fix the mathematics before recruitment expands.
- If users interpret two checks as permanent mastery, change the language and progress display.
- If teacher access or reporting fails, narrow the pilot to a workflow the partner can actually sustain.
- If route compression confuses or pressures learners, simplify it and compare it with ordinary progress feedback.

These are qualitative improvement rules, not statistical thresholds. Preserve negative observations and count all tested sessions, including abandoned ones.

How findings change the application

Replace “we plan to interview” with the actual number, date range, recruitment method, and a restrained finding only after the sessions occur. Report exceptions. For example, use “In 12 permitted walkthroughs, 5 participants needed help locating the practice link; we moved it next to the lesson.” Do not use this example as a result.

If no validation occurs before the deadline, submit the honest pre-pilot account. Do not insert fictional quotes or imply the team observed a barrier it has only inferred. Keep a later validation update for nationals.

Final pre-submit check

Confirm the institution's legal name and campus, NCR region, coach's formal faculty role and department, all students' programs/year levels/emails, required contact numbers, campus exclusivity, and consent to submit. Justin Mesias is supplied as coach; the team must confirm that he meets the organizer's faculty-adviser requirement.

The live form describes 300 words for all seven narratives and a maximum 15-page PDF. Its executive-summary field additionally contains `max_length 300`, which appears to impose a character limit. Test the field during authorized form completion or ask the organizer for correction. Do not bypass validation or submit invented identity details to inspect later pages. Use the prepared short fallback if necessary.